


 Name: For ANSWERS

Total Marks: 40 marks

Total Time: 45 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Resources Allowed: 1 page A4 notes (1 side), calculator.

Question 1**11 marks**

Calculate the mean, median, mode and range for the following set of data.

23, 32, 16, 37, 15, 31, 2, 25, 2, 15, 16, 23, 25, 31, 32, 37

	With outlier	Without outlier
Mean	22.625 ✓	22.25 25.57 ✓
Median	24 ✓	25 ✓
Mode	No mode ✓	No mode ✓
Range	35 ✓	22 ✓

a) Identify the outlier in the data: 2 ✓

b) What effect does the outlier have on the data?

mean & median average has increased ✓
 Range of data decreased (makes it less spread out) ✓

Question 2**3 marks**

What type of data are the following?

Choose from categorical, discrete numerical or continuous numerical:

a) the number of hours spent playing video games each week

discrete numerical

b) the favourite pizza topping of each of the students in a class

categorical

c) the length of time that a year 8 student can spend running

continuous numerical

Question 3

7 marks

Give one advantage and disadvantage of conducting a census and sample

	Census	Sample
Advantage	more accurate data	quick & easy
Disadvantage	expensive	less accurate data.

Classify each of the following data collections according to whether they are a sample or census

- a) You want to know the favourite sport to watch in Perth, so you ask every student and Baldivis Secondary College

sample

- b) Number of hours spent completing homework each week by your class by surveying each student in your class.

census

- c) Asking every student that walks through the gates of Baldivis Secondary College how they travel to school.

census

Question 4

2 marks

Use only the numbered cards below to answer the following questions.



Choose 3 numbers so that:

- a) no mode range = 6 mean = 6

3 6 9 ✓

Choose 4 numbers so that:

- b) mean = 4 mode = 4

1 4 4 7 ✓

Question 5

10 marks

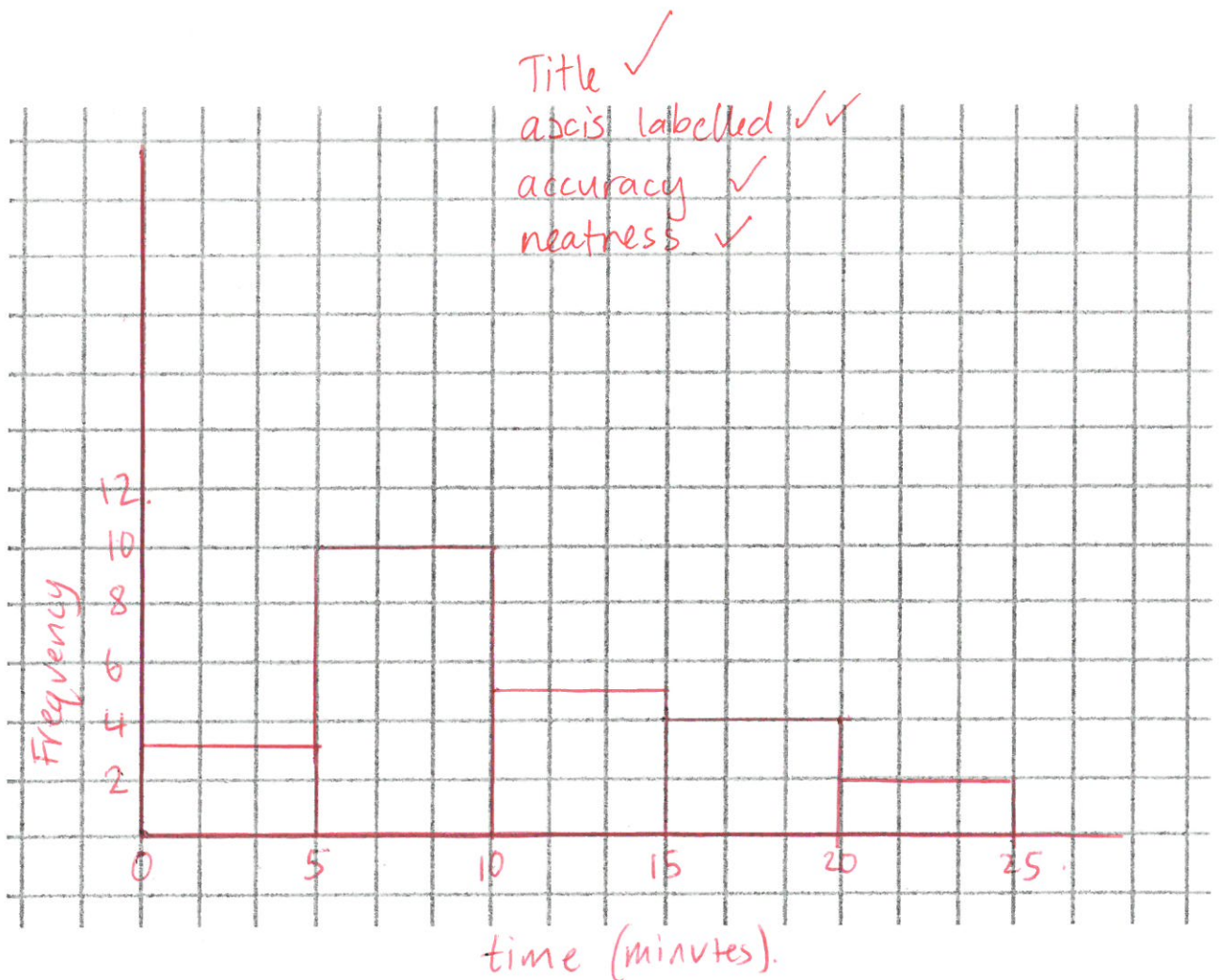
The times taken by 24 students to complete a mathematical puzzle are shown below in minutes:

~~5, 22, 8, 13, 7, 9, 6, 8, 12, 10, 4, 9, 3, 19, 18, 9, 14, 2, 5, 15, 21, 11, 5, 17~~

a) Create a grouped frequency table

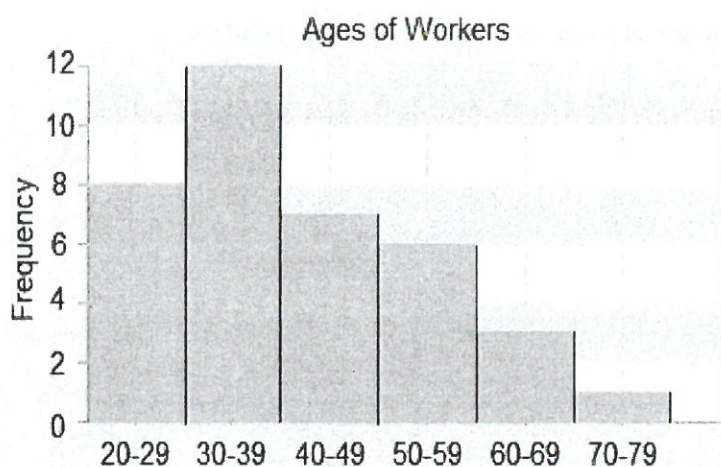
Time taken (minutes)	Tally	Frequency
0 - 4		3 ✓
5 - 9		10 ✓
10 - 14		5 ✓
15 - 19		4 ✓
20 - 24		2 ✓

b) Construct a histogram using the data above



Question 6

3 marks



a) Would you describe the shape as symmetrical or skewed?

skewed

b) How many workers were over the age of 60?

4

c) How many workers were interviewed in total?

37.

Question 7

4 marks

Students were asked to find the mean, median, mode and range for the Stem and Leaf plot. Tom gave the following answers. **CIRCLE** which of his answers are incorrect? AND give the correct answer showing working out.

stem	leaf
2	1, 5, 6
3	0, 0, 1
4	0

Mean = 29 ✓

median = 26

mode = 30 ✓

range = 34

median = 30 ✓

~~21~~, ~~25~~, ~~26~~, 30, ~~30~~, ~~31~~, ~~40~~

range = 19. ✓

$40 - 21 = \underline{19}$

END OF TEST ☺